

MSc Stereo Depth Estimation Project

This repository contains the experimental code and selected results for my MSc dissertation project.

****Depth Estimation using Dual-Camera Systems: A Comprehensive Survey and Comparative Study****

The project investigates dual-camera / stereo depth estimation methods, combining a structured :

Project Overview

Depth estimation from dual-camera systems is a fundamental problem in computer vision and is widely studied.

This project focuses on stereo depth estimation methods and evaluates representative classical :

- KITTI Stereo 2015 data loading
- Ground-truth disparity reading and visualisation
- StereoBM baseline
- Semi-Global Block Matching baseline
- Quantitative evaluation using EPE, Bad-3px, runtime, and valid prediction ratio
- Qualitative disparity map comparison
- SGBM block-size parameter sensitivity analysis

The broader dissertation also discusses deep stereo, lightweight stereo, and mobile dual-camera

Repository Structure

```text

msc-stereo-depth-estimation/

|

├─ src/

|

├─ utils.py

|

├─ check\_kitti.py

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├─ run\_bm\_sgbm\_kitti.py

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├─ run\_sgbm\_parameter\_study.py

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└─ summarize\_results.py

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├─ outputs/

|

├─ metrics/

```
| | | └─ bm_sgbm_kitti_results.csv
| | | └─ sgbm_blocksize_study_detailed.csv
| | |
| | └─ summaries/
| | └─ average_results.csv
| | └─ sgbm_blocksize_study_summary.csv
| |
| | └─ figures/
| | └─ selected_examples/
| | └─ sgbm_parameter_study/
|
└─ docs/
└─ configs/
└─ requirements.txt
└─ .gitignore
└─ LICENSE
└─ README.md
```

## Dataset

The experiments use the **KITTI Stereo 2015** dataset.

The dataset is not included in this repository due to file size and dataset licensing considerations. Users should download the dataset from the official KITTI website and place it locally using the following structure:

```
data/KITTI_2015/
 training/
 image_2/
 image_3/
 disp_occ_0/
 testing/
```

In my local setup, the dataset path is provided using the environment variable:

```
export KITTI_2015_ROOT=/path/to/KITTI_2015
```

For example:

```
export KITTI_2015_ROOT=/home/milky/msc_stereo_depth/data/KITTI_2015
```

# Environment

The project was developed using:

- Windows + WSL2
- Ubuntu
- Conda
- Python
- OpenCV
- NumPy
- Matplotlib

Install the required Python packages with:

```
pip install -r requirements.txt
```

# How to Run

## 1. Check KITTI Loading

```
python src/check_kitti.py
```

This script checks whether the left image, right image, and ground-truth disparity map can be loaded correctly.

Expected output includes:

```
KITTI sample loaded successfully.
check_kitti.py completed successfully.
```

## 2. Run StereoBM and SGBM Baseline Experiment

```
python src/run_bm_sgbm_kitti.py
```

This script runs StereoBM and SGBM on a selected number of KITTI image pairs and saves:

```
outputs/metrics/bm_sgbm_kitti_results.csv
outputs/figures/
outputs/figures/selected_examples/
```

The current baseline experiment uses 20 KITTI samples.

## 3. Summarise Baseline Results

```
python src/summarize_results.py
```

This generates the average result summary:

```
outputs/summaries/average_results.csv
```

The summary includes:

- Average EPE
- Average Bad-3px error rate
- Average runtime
- Average valid prediction ratio

## 4. Run SGBM Block-Size Parameter Study

```
python src/run_sgbm_parameter_study.py
```

This script evaluates SGBM using different block sizes:

```
blockSize = 3, 5, 7, 9, 11
```

The experiment uses 20 KITTI samples and saves:

```
outputs/metrics/sgbm_blocksize_study_detailed.csv
outputs/summaries/sgbm_blocksize_study_summary.csv
outputs/figures/sgbm_parameter_study/
```

## Evaluation Metrics

The project uses the following metrics:

### End-Point Error

End-Point Error measures the average absolute disparity error over valid pixels.

### Bad-3px Error Rate

Bad-3px measures the percentage of pixels where the disparity error is greater than 3 pixels. Invalid predictions over valid ground-truth regions are also treated as bad pixels.

### Runtime

Runtime is measured in milliseconds per stereo image pair.

### Valid Prediction Ratio

Valid prediction ratio measures the percentage of ground-truth-valid pixels where the method also produces a valid disparity prediction.

## Baseline Experiment

The baseline experiment compares two classical stereo matching methods on 20 KITTI Stereo 2015 samples:

| Method   | Samples | Average EPE | Bad-3px (%) | Runtime (ms) | Valid Ratio (%) |
|----------|---------|-------------|-------------|--------------|-----------------|
| StereoBM | 20      | 1.586       | 59.08       | 18.49        | 43.07           |
| SGBM     | 20      | 1.300       | 17.62       | 46.05        | 87.24           |

The results show that SGBM substantially outperforms StereoBM in terms of disparity quality and prediction coverage. Compared with StereoBM, SGBM reduces the average EPE from 1.586 to 1.300 and lowers the Bad-3px error rate from 59.08% to 17.62%. It also increases the valid prediction ratio from 43.07% to 87.24%, indicating that SGBM produces valid disparity estimates over a much larger portion of the ground-truth-valid regions.

However, this improvement comes with a higher computational cost. The average runtime increases from 18.49 ms for StereoBM to 46.05 ms for SGBM. This demonstrates a clear accuracy-robustness-runtime trade-off: SGBM provides more reliable and spatially coherent disparity maps, while StereoBM is faster but produces more fragmented and incomplete predictions.

Initial findings show that SGBM generally produces more spatially coherent disparity maps and a

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## ## SGBM Block-Size Parameter Study

The SGBM block-size sensitivity experiment evaluates how the local matching window size affects

| blockSize | Average EPE | Bad-3px (%) | Runtime (ms) | Valid Ratio (%) |
|-----------|-------------|-------------|--------------|-----------------|
| -----:    | -----:      | -----:      | -----:       | -----:          |
| 3         | 1.442       | 19.18       | 25.84        | 86.44           |
| 5         | 1.300       | 17.62       | 25.09        | 87.24           |
| 7         | 1.251       | 17.04       | 36.76        | 87.53           |
| 9         | 1.273       | 16.96       | 52.90        | 87.54           |
| 11        | 1.350       | 17.30       | 59.44        | 87.35           |

The results show that increasing the block size from 3 to 7 improves the average EPE, suggesting

Overall, `blockSize = 7` gives the lowest EPE, while `blockSize = 5` provides a strong speed-ac

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## ## Example Qualitative Results

Selected qualitative examples are stored in:

```
```text
outputs/figures/selected_examples/
```

These figures compare:

- Left image
- Ground-truth disparity
- StereoBM estimated disparity
- SGBM estimated disparity

The qualitative results show that StereoBM often produces fragmented disparity maps with many invalid regions, especially in weak-texture areas and around object boundaries. SGBM produces smoother and more complete disparity maps, but still struggles with occlusion boundaries, thin structures, reflective vehicle surfaces, and close-range objects.

Current Progress

- ☒ KITTI Stereo 2015 loading
- ☒ Ground-truth disparity visualisation
- ☒ StereoBM baseline
- ☒ SGBM baseline
- ☒ 20-sample baseline experiment
- ☒ Quantitative result summary
- ☒ Qualitative comparison figures
- ☒ SGBM block-size parameter sensitivity analysis
- ☐ Dissertation methodology writing
- ☐ Dissertation results analysis
- ☐ Optional additional parameter study
- ☐ Optional deep stereo model comparison

Dissertation Relevance

This repository supports the experimental component of the MSc dissertation. The results will be used to discuss:

- The difference between local and semi-global stereo matching
- Accuracy-runtime trade-offs in classical stereo methods
- Failure cases in weak-texture, occlusion, and object-boundary regions
- The practical limitations of stereo depth estimation for mobile and dual-camera systems

The experimental findings will be connected with the literature review on classical stereo, deep stereo, lightweight stereo, and mobile dual-camera depth estimation.

Notes

The KITTI dataset is not included in this repository. Only code, selected outputs, and lightweight result summaries are included.

